

daig

• Intern Cheat Card

Break it. Diagnose it. Ship it safely.

github.com/adeelarshad414/devops-interns-proj · MIT · runs on a laptop



● Run the stack

```
make up                # the 3 tiers
make seed              # data
make smoke             # verify tiers
make psql              # db shell
make down / nuke      # stop / +wipe
```

● Observe & diagnose

```
make obs              # Grafana :3000
make load             # iftar spike
```

1 Metric p95 over the 1s SLO?

2 Trace where is the time?

3 Log what happened? (trace_id)

4 Profile which line? (hot loop)

● Localhost

```
web    http://localhost:8080
grafana http://localhost:3000
prometheus http://localhost:9090
openbao http://localhost:8200
```

● Kubernetes

```
kubectl apply -k k8s/base
kubectl -n daig get pods -w
kubectl -n daig get hpa
kubectl -n daig rollout undo \
deploy/orders
kubectl -n daig logs <p> --previous
```

● IaC — Terraform & Ansible

```
terraform -chdir=infra/aws \
init -backend=false && validate
ansible-playbook -i ../hosts.yml \
site.yml                                # run twice
```

● CI/CD

Break a test → the build refuses.

Immutable :sha; :latest only on main.

CD: guard → canary → rollout → rollback.

● The 7 phases

P0-P1 baseline · run · exit-78

P2-P3 observe · SLO · k8s · rollback

P4-P5 IaC · CI/CD refuses

P6-P7 DevSecOps · harden

→ leave evidence + a gate each phase.

● Security & Secrets

```
make scan                # full toolchain
make scan-sast           # fast
make insecure-on         # planted vulns
./chaos/day6-security.sh verify
make vault-up && make vault-app
```

● Break it (chaos)

```
./chaos/day1-network.sh break|fix
./chaos/day3-crashloop.sh break|fix
./chaos/day4-latency.sh break|fix
```

One fault. Know the revert first.

Nothing alerts? That’s the bug.

● Checks (no Docker)

```
make check              # js·shell·yaml·json
make test               # unit tests
make hooks              # pre-commit gate
```

● Golden rules

- Fail loud, fail closed
- Reproducible or it didn’t happen
- :latest is not a version
- Least privilege everywhere
- Health ≠ readiness ≠ liveness
- One owner per field (HPA owns replicas)
- Alert on symptoms, page on burn rate
- Gate what you fix